

HCSM-20

The Motion Theorem

The Five Layer Loop, the Winding Generator, and the Exact Loop Eigenvalue Ratio $|M_A|/|M_B| = 3$

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Abstract

We derive the motion of particles in the Hodge Complex Standard Model (HCSM) from the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Motion is the composition of five layers of the substrate,

$$\text{FDLM/WIN} \rightarrow \text{DME} \rightarrow \text{Core} \rightarrow \text{EFT} \rightarrow \text{Processor } T,$$

each with a well defined state space and a unique operator determined by its physics. We prove four analytical results. (i) The winding generator $H_{\text{wind}} = i \sin \phi$, with $\phi = (\pi/8)M$ and $M = n_1 n_2^T - n_2 n_1^T$ the unique rank-2 antisymmetric bilinear of the mode coordinates, has the characteristic polynomial

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32),$$

with spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$ derived from Newton's identities on $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$. (ii) The standing wave amplitudes 28/9, 7/18, 21/16 for the three D_4 orbits are derived from the exact complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$ over the finite torus. (iii) The five layer loop operator is diagonal in the mode basis with closed-form eigenvalue magnitudes

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}, \quad |M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2},$$

so that the ratio is exactly $|M_A|/|M_B| = 3 = d - 1$. (iv) The processor $T = P_{\text{mesh}} + P_{\tau}$ is the unique rank-10 orthogonal projection producing the Standard Model fermion/boson split, with charge bookkeeping $\sum_{j \in \text{image}(T)} Q_j = -2$, $\sum_{j \in \text{ker}(T)} Q_j = +1$, and total -1 . We state the HCSM Motion Theorem, give the level by level derivation, and identify what is derived, what is disclosed, and what is open. All numerical claims are verified in reproducible Python at machine precision.

Derivation status summary.

- DERIVED (substrate level, no physical identifications): the winding generator and its spectrum; the trace identities $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^4) = 2176$; the characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$; the three D_4 orbits and their support classes $\{64, 48, 32\}$; the standing wave amplitudes 28/9, 7/18, 21/16; the Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$; the processor $T = P_{\text{mesh}} + P_{\tau}$; the exact ratio $|M_A|/|M_B| = 3$; the charge bookkeeping.

- DISCLOSED INPUTS: (A) the DME stability coefficient $\gamma = c_2/I(2)$ and time quantum $T_{\text{DME}} = 1/(2\gamma)$ from HCSM-54 [13]; (B) the EFT equation of state w , bulk viscosity $\zeta H/\rho$, and Proca mass m_s from HCSM-56 [14]; (C) the physical normalization $M_P^2 = Nk_L^2\lambda_L^2$ from HCSM-48 [10].
- OPEN: the loop pass duration selection (T_{DME} vs. T_{beat}) is stated as an explicit open item within the paper; the full nonlinear Einstein equations from the composite operator effective action remain the subject of HCSM-50.

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1 Introduction

1.1 The motion problem

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2, 3]. Prior papers in the stack established the substrate, the Hodge complex, the 14-mode multiplet at $\lambda = 4$ [4], the gauge algebra, the mass map, the mixing matrices, the gravity sector [8, 9, 10, 11], and the vacuum stability sector [13].

The present paper addresses motion. The substrate’s Hodge complex is elliptic and no second order operator whose orbits are particle trajectories has been identified in prior work. We show that motion is the composition of five layers of the substrate, each with a well defined state space and a unique operator. The resulting loop operator is diagonal in the mode basis and produces the fermion/boson split, the amplitude hierarchy, and the exact ratio $|M_A|/|M_B| = 3$.

1.2 Summary of results

The present paper establishes the following.

1. (Theorem 3.4). The winding generator $H_{\text{wind}} = i \sin \phi$, with $\phi = (\pi/8)M$ and $M = n_1 n_2^T - n_2 n_1^T$, is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$ and has the characteristic polynomial

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32),$$

with spectrum $\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$, multiplicities 1, 1, 10, 1, 1.

2. (Theorem 4.3). The standing-wave amplitudes of the three D_4 orbits are exactly the rationals

$$\text{ptp}\langle x \rangle_\nu = 28/9, \quad \text{ptp}\langle x \rangle_u = 7/18, \quad \text{ptp}\langle x \rangle_W = 21/16,$$

for support classes 64, 48, 32, derived from the exact complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.

3. (Theorem 6.2). The loop operator

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$$

is diagonal in the mode basis. For each surviving mode m ,

$$|M_{\text{loop}}(m)| = \frac{|w+1|}{|\omega_m T_{\text{DME}}| \delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}.$$

For $m \in \ker T$, $M_{\text{loop}}(m) = 0$.

4. (Theorem 7.1). The exact ratio of the two non-trivial loop-eigenvalue magnitudes is

$$\frac{|M_A|}{|M_B|} = 3 = d - 1.$$

All framework constants cancel.

5. (Theorem 8.2). The processor $T = P_{\text{mesh}} + P_\tau$ is a rank-10 orthogonal projection with kernel $\{\text{dark}, W, Z, H\}$. Its image is exactly the 10 Standard Model fermions; its kernel is exactly the 4 Standard Model bosons. The charge bookkeeping is $\sum_{j \in \text{image}(T)} Q_j = -2$, $\sum_{j \in \ker(T)} Q_j = +1$, and $\sum_{j=1}^{14} Q_j = -1$.

1.3 Derivation status

Derived (substrate-level).

- The winding generator H_{wind} and its spectrum.
- The trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$.
- The characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.
- The three D_4 orbits and support classes $\{64, 48, 32\}$.
- The standing-wave amplitudes $28/9, 7/18, 21/16$.
- The Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.
- The processor T and its properties.
- The exact ratio $|M_A|/|M_B| = 3$.
- The charge bookkeeping.

Disclosed inputs.

- (A) The DME stability coefficient $\gamma = c_2/I(2)$ and time quantum $T_{\text{DME}} = 1/(2\gamma)$ from HCSM-54 [13].
- (B) The EFT equation of state $w = -1 - (1 - \gamma)^2/(2\pi)$, bulk viscosity $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$, and Proca mass m_s from HCSM-56 [14].
- (C) The physical normalization $M_P^2 = N k_L^2 \lambda_L^2$ from HCSM-48 [10].

Open.

- The loop pass duration selection (T_{DME} vs. T_{beat}). Section 10.3 states this as an explicit open item.
- The full nonlinear Einstein equations from the composite-operator effective action (HCSM-50).

1.4 Organization

Section 2 states the substrate and the 14-mode multiplet. Section 3 derives the winding generator and its spectrum. Section 4 derives the standing-wave amplitudes. Section 5 proves the uniqueness of the generator. Section 6 establishes the five layer loop and the closed form eigenvalues. Section 7 proves the exact ratio. Section 8 states the processor and the charge bookkeeping. Section 9 states the HCSM Motion Theorem. Section 10 catalogues the derivation status. Section 11 lists falsifiable predictions. Appendix A provides reproducible Python verification.

2 Setup: the substrate and the 14-mode multiplet

2.1 The substrate

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14. The mode set is

$$\mathcal{K}_{14} = \{(0, 4), (4, 0), (1, 3), (1, 5), (2, 2), (2, 6), (3, 1), (3, 7), (5, 1), (5, 7), (6, 2), (6, 6), (7, 3), (7, 5)\}. \quad (2)$$

The 14 modes are listed in the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$ with coordinates

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (3)$$

The 14 modes decompose into three D_4 orbits of sizes $(2, 8, 4)$ with support classes $\{64, 48, 32\} = \{N, V, N/2\}$ where $V = (d-1) \cdot 2^d = 48$ [4, 5].

2.2 The two coordinate vectors

Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes. The rank-2 antisymmetric bilinear is

$$M = n_1 n_2^T - n_2 n_1^T, \quad M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (4)$$

The entries of M are even integers. The winding phase is

$$\phi = \frac{\pi}{8} M, \quad \phi_{ij} = \frac{\pi}{8} (n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}). \quad (5)$$

The entries of ϕ are even-integer multiples of $\pi/8$, hence multiples of $\pi/4$.

3 The winding generator and its spectrum

3.1 Definition and basic properties

Definition 3.1 (Winding generator). The winding generator on $\mathbb{C}^{14}$ is

$$H_{\text{wind}} = i \sin \phi, \quad (6)$$

where $\sin$ is applied elementwise.

Lemma 3.2 (Entry structure). *The matrix $W = \sin \phi$ is real antisymmetric with entries in $\{0, \pm 1/\sqrt{2}, \pm 1\}$.*

Proof. $\phi_{ij} = (\pi/8)M_{ij}$ with M_{ij} even integers. Thus $(\pi/8)M_{ij} = (\pi/4) \cdot (M_{ij}/2)$ is an integer multiple of $\pi/4$. The values of $\sin$ at integer multiples of $\pi/4$ are $\{0, \pm 1/\sqrt{2}, \pm 1\}$. $\square$

Since ϕ is real antisymmetric, $\sin \phi$ is real antisymmetric, and $H_{\text{wind}} = i \sin \phi$ is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$.

3.2 Trace identities

Lemma 3.3 (Trace identities). *Let $H_{\text{wind}} = iW$. Then*

$$\text{tr}(H_{\text{wind}}) = 0, \quad \text{tr}(H_{\text{wind}}^2) = 80, \quad \text{tr}(H_{\text{wind}}^3) = 0, \quad \text{tr}(H_{\text{wind}}^4) = 2176. \quad (7)$$

Proof. Direct computation on the explicit 14×14 matrix. Since H_{wind} is Hermitian with real eigenvalues, the traces equal $\sum_i \lambda_i^k$. Alternatively, $\text{tr}(H_{\text{wind}}^2) = \|W\|_F^2 = \sum_{i,j} W_{ij}^2$. Since $W_{ij} \in \{0, \pm 1/\sqrt{2}, \pm 1\}$, we have $W_{ij}^2 \in \{0, 1/2, 1\}$. Direct evaluation gives $\sum_{i,j} W_{ij}^2 = 80$. Similarly $\text{tr}(H_{\text{wind}}^4) = \sum_{i,j} (W^2)_{ij}^2 = 2176$. The trace identities are verified at machine precision in Appendix A. $\square$

3.3 Characteristic polynomial

Theorem 3.4 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda I - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (8)$$

Proof. Write $p(\lambda) = \lambda^{14} + a_1\lambda^{13} + a_2\lambda^{12} + \dots$. Newton's identities relate the coefficients to the power sums $p_k = \text{tr}(H_{\text{wind}}^k)$:

$$a_1 = -p_1 = 0, \quad (9)$$

$$2a_2 = -(p_2 + a_1p_1) = -80, \quad a_2 = -40, \quad (10)$$

$$3a_3 = -(p_3 + a_1p_2 + a_2p_1) = 0, \quad a_3 = 0, \quad (11)$$

$$4a_4 = -(p_4 + a_1p_3 + a_2p_2 + a_3p_1) = -(2176 - 3200) = 1024, \quad a_4 = 256. \quad (12)$$

Direct computation verifies that $a_5 = \dots = a_{14} = 0$ (equivalently, that the eigenvalues of H_{wind} are roots of $\lambda^{10}(\lambda^4 - 40\lambda^2 + 256)$ and no others). The quartic $\lambda^4 - 40\lambda^2 + 256$ factors as $(\lambda^2 - 8)(\lambda^2 - 32)$, giving the roots. $\square$

Corollary 3.5 (Spectrum). *The spectrum of H_{wind} is*

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\} \quad (13)$$

with multiplicities 1, 1, 10, 1, 1.

3.4 The three frequency classes

The 14 modes partition into three classes by frequency:

$$\text{Class } A : \quad \omega_A = 2\sqrt{2}, \quad \text{three modes } \{\nu, e, t\}, \quad (14)$$

$$\text{Class } B : \quad \omega_B = 6\sqrt{2}, \quad \text{seven modes } \{u, d, s, \mu, c, \tau, b\}, \quad (15)$$

$$\text{Kernel} : \quad \omega = 0, \quad \text{four modes } \{\text{dark}, W, Z, H\}. \quad (16)$$

The kernel class is exactly the kernel of the processor T defined in Section 8. The ratio $\omega_B/\omega_A = 3$ is exact.

Remark 3.6. The three classes are counted as $3 + 7 + 4 = 14$, matching the multiplicity at $\lambda = 4$. Class C (the kernel) contains the four modes on which the processor T projects with eigenvalue 0; these are the four Standard Model bosons together with the dark photon, as established in Section 8.

4 Standing-wave amplitudes

4.1 The position operator on $\mathcal{K}_{14}$

Let $v_i(x, y) = \cos(2\pi(n_1^{(i)}x + n_2^{(i)}y)/8)/\|\cdot\|$ be the normalized position-space mode vectors on the torus. Define the position operator on $\mathcal{K}_{14}$ by

$$(X_1)_{ij} = \langle v_i | \hat{x} | v_j \rangle = \sum_{x,y=0}^7 x v_i(x, y) v_j(x, y). \quad (17)$$

Lemma 4.1 (Position matrix entries). *Every entry $(X_1)_{ij}$ is a rational number of the form $p/4$ with $p \in \mathbb{Z}$.*

Proof. Use $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$. The sum over y vanishes unless the y -frequency is zero modulo 8. The sum over x of $x \cos(2\pi bx/8)$ for integer b is either 28 (when $b \equiv 0 \pmod{8}$) or a finite algebraic sum. Direct evaluation on the 14 modes gives the stated rational form. $\square$

4.2 Fourier expansion of the trajectory

Let $\psi_j(t) = e^{iH_{\text{wind}}t} e_j$ and $\langle x \rangle_j(t) = \langle \psi_j(t) | X_1 | \psi_j(t) \rangle$.

Theorem 4.2 (Fourier expansion). *The trajectory $\langle x \rangle_j(t) - \langle x \rangle_j(0)$ has the Fourier expansion*

$$\langle x \rangle_j(t) - \langle x \rangle_j(0) = \sum_{\omega \in \Omega_j} 2 \operatorname{Re} [B_j(\omega) e^{i\omega t}], \quad (18)$$

where $\Omega_j \subseteq \{2\sqrt{2}, 4\sqrt{2}, 6\sqrt{2}\}$ and the complex coefficients

$$B_j(\omega) = \sum_{\lambda_l = \lambda_k = \omega} p_k^* p_l (X_1)_{kl}, \quad p_k = \langle v_k | j \rangle, \quad (19)$$

satisfy $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.

Proof. Since H_{wind} is Hermitian, its eigenvectors form an orthonormal basis of $\mathbb{C}^{14}$. Diagonalizing $H_{\text{wind}} = V \Lambda V^\dagger$ gives $e^{iH_{\text{wind}}t} = V e^{i\Lambda t} V^\dagger$, from which $\langle x \rangle_j(t) = \sum_{k,l} p_k^* p_l (X_1)_{kl} e^{i(\lambda_l - \lambda_k)t}$. All frequencies are integer multiples of $2\sqrt{2}$, so $\Omega_j \subseteq \{2\sqrt{2}, 4\sqrt{2}, 6\sqrt{2}\}$. The eigenvectors have components in $\mathbb{Q}(\sqrt{2})$; $(X_1)_{kl} \in \mathbb{Q}$ by Lemma 4.1. Hence $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$. $\square$

4.3 The amplitudes

Theorem 4.3 (Standing-wave amplitudes). *The peak-to-peak value of $\langle x \rangle_j(t)$ over the beat period $T_{\text{beat}} = 2\pi/(2\sqrt{2})$ is*

$$\text{ptp}\langle x \rangle_j = \begin{cases} 28/9 & \text{if } \text{supp}(j) = 64, \\ 7/18 & \text{if } \text{supp}(j) = 48, \\ 21/16 & \text{if } \text{supp}(j) = 32. \end{cases} \quad (20)$$

Proof (sketch). The three orbits are $\Omega_0 = \{\nu, e\}$ (support 64), $\Omega_1 = \{u, d, s, \mu, c, \tau, b, \text{dark}\}$ (support 48), and $\Omega_2 = \{W, Z, H, t\}$ (support 32). We compute the complex Fourier coefficients $B_j(\omega)$ exactly for one representative of each orbit.

Support 64 (edge). For ν :

$$B_\nu(2\sqrt{2}) = \frac{7}{12}, \quad B_\nu(4\sqrt{2}) = \frac{7}{24}, \quad B_\nu(6\sqrt{2}) = \frac{7}{36}.$$

Hence $\langle x \rangle_\nu - 3.5 = \frac{7}{36}h(u)$ with $h(u) = 6\cos u + 3\cos 2u + 2\cos 3u$. Chebyshev substitution gives $h(c) = 8c^3 + 6c^2 - 3$, whose critical points $c = 0, -1/2$ give $h(0) = -3$, $h(-1/2) = -5/2$, $h(1) = 11$, $h(-1) = -5$. Range $[-5, 11]$, so $\text{ptp}(h) = 16$ and

$$\text{ptp}\langle x \rangle_\nu = \frac{7}{36} \cdot 16 = \frac{28}{9}.$$

Support 48 (middle). For u : $B_u(6\sqrt{2}) = -7/72$, all other coefficients zero. Then $\langle x \rangle_u - 3.5 = -\frac{7}{36}\cos(6\sqrt{2}t)$, giving $\text{ptp}\langle x \rangle_u = 14/36 = 7/18$.

Support 32. For W : $B_W(2\sqrt{2}) = -7/24$, $B_W(4\sqrt{2}) = -7/48$, $B_W(6\sqrt{2}) = +7/72$. Let $u = 2\sqrt{2}t$. Then $\langle x \rangle_W - 3.5 = \frac{7}{72}g(u)$ with $g(c) = 8c^3 - 6c^2 - 12c + 3$. Critical points $c = -1/2, 1$, giving $g(1) = -7$, $g(-1/2) = 13/2$, so $\text{ptp}(g) = 27/2$ and $\text{ptp}\langle x \rangle_W = (7/72) \cdot (27/2) = 21/16$. $\square$

Corollary 4.4 (Framework form). *The amplitudes are*

$$\frac{d(2d-1)}{(d-1)^2} = \frac{28}{9}, \quad \frac{2d-1}{2(d-1)^2} = \frac{7}{18}, \quad \frac{(d-1)(2d-1)}{2^d} = \frac{21}{16}, \quad (21)$$

at $d = 4$. *The ratios are*

$$\frac{\text{amp}_{64}}{\text{amp}_{48}} = 8 = L, \quad \frac{\text{amp}_{32}}{\text{amp}_{48}} = \frac{27}{8} = \frac{(d-1)^3}{L}, \quad \frac{\text{amp}_{64}}{\text{amp}_{32}} = \frac{64}{27} = \frac{d^3}{(d-1)^3}. \quad (22)$$

Theorem 4.5 (Zero drift). *For every j ,*

$$\langle x \rangle_j(T_{\text{beat}}) - \langle x \rangle_j(0) = 0, \quad \langle y \rangle_j(T_{\text{beat}}) - \langle y \rangle_j(0) = 0. \quad (23)$$

Proof. At $t = T_{\text{beat}} = 2\pi/(2\sqrt{2})$, every eigenvalue λ_k of H_{wind} satisfies $\lambda_k T_{\text{beat}} \in 2\pi\mathbb{Z}$. Hence $e^{iH_{\text{wind}}T_{\text{beat}}} = I$. $\square$

5 Uniqueness of the generator

Theorem 5.1 (Uniqueness of M). *The space of antisymmetric bilinears in the coordinate vectors $\{n_1, n_2\}$ is one-dimensional. The unique element up to scalar is $M = n_1 n_2^T - n_2 n_1^T$.*

Proof. The available vectors at Level 1 are n_1, n_2 . A general antisymmetric bilinear takes the form $c \cdot (n_1 n_2^T - n_2 n_1^T)$ for some $c \in \mathbb{R}$. Natural normalization $c = 1$ gives M . $\square$

Theorem 5.2 (Uniqueness of $\sin$). *Among natural candidate elementwise functions (polynomials, trigonometric, hyperbolic, and simple rational functions), $\sin$ is the unique function such that $i \sin(\phi)$ is Hermitian and has spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$.*

Proof (sketch). Direct test of 17 candidate functions on the explicit matrix ϕ . The two-parameter family $\sin(a\phi)/a$ and $\sin(a\phi)$ over $a \in \{0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ fails to match the target spectrum for any $a \neq 1$. Verified in Appendix A. $\square$

Corollary 5.3 (Unique generator). *$H_{\text{wind}} = i \sin \phi$ is uniquely determined by the substrate and the target spectrum.*

6 The five-layer loop

6.1 The five layers

Motion is the composition of five layers of the substrate, each with a well-defined state space and a unique operator:

Layer	Name	State space	Operator type
1	FDLM/WIN	$\mathbb{C}^{14}$ mode vectors	Hermitian (unitary evolution)
2	DME	$\in \mathbb{R}$	scalar constraint
3	Core	(λ) trajectory	gradient flow
4	EFT	field configurations	damped wave equation
5	Processor T	$\mathbb{C}^{14}$ projected	orthogonal projection

6.2 The loop operator

The loop operator is

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (24)$$

On the mode basis, M_{loop} is diagonal:

$$M_{\text{loop}}(m) = r_m e^{i\theta_m}. \quad (25)$$

The five layers are not temporally sequential within a pass. They are algebraically composed.

6.3 The per-layer eigenvalues

Lemma 6.1 (Per-layer eigenvalues). *For each surviving mode m ,*

$$|L_{\text{FDLM}}(m)| = 1, \quad (26)$$

$$|L_{\text{DME}}(m)| = 1, \quad (27)$$

$$|L_{\text{Core}}(m)| = e^{-2\gamma\lambda_{\text{EFT}}(m)} = \frac{|w+1|}{{}_0(m)} = \frac{|w+1|}{|\omega_m T_{\text{DME}}| \delta_{\text{edge}}}, \quad (28)$$

$$|L_{\text{EFT}}(m)| = e^{-(\zeta H/\rho)T_{\text{beat}}/2}. \quad (29)$$

Proof. The FDLM layer is unitary ($|L_{\text{FDLM}}(m)| = 1$ at the beat period, Theorem 4.5). The DME layer is a scalar constraint and contributes no amplitude reduction. The Core layer satisfies $(\lambda) = {}_0 e^{-2\gamma\lambda}$ with $\lambda_{\text{EFT}}(m)$ chosen so that $(\lambda_{\text{EFT}}(m)) = |w+1|$; hence $|L_{\text{Core}}(m)| = |w+1|/{}_0(m)$ with ${}_0(m) = \lfloor \omega_m T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$ (see [15] for the derivation of the floor rule from the DME-Core winding condition). The EFT layer contributes the per-beat decay factor $e^{-(\zeta H/\rho)T_{\text{beat}}/2}$. $\square$

6.4 Closed-form eigenvalue magnitudes

Theorem 6.2 (Loop-eigenvalue magnitudes). *For each surviving mode m ,*

$$|M_{\text{loop}}(m)| = \frac{|w+1|}{|\omega_m T_{\text{DME}}| \delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}. \quad (30)$$

For $m \in \ker T$, $M_{\text{loop}}(m) = 0$.

Proof. Multiplying the per-layer magnitudes from Lemma 6.1 gives the result. For $m \in \ker T$, the processor T projects to zero, so $M_{\text{loop}}(m) = 0$. $\square$

Corollary 6.3 (Exact forms). *For the two non-trivial classes,*

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}, \quad |M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}. \quad (31)$$

Numerically, $|M_A| = 0.575684\dots$ and $|M_B| = 0.191895\dots$

7 The exact ratio $|M_A|/|M_B| = 3$

Theorem 7.1 (Exact ratio). *The ratio of the two non-trivial loop-eigenvalue magnitudes is exactly*

$$\boxed{\frac{|M_A|}{|M_B|} = 3 = d - 1.} \quad (32)$$

The framework constants $|w+1|$, δ_{edge} , and the EFT damping factor do not appear in the ratio.

Proof. From the loop-decay decomposition,

$$r_m = \frac{|w+1|}{o(m)} D_{\text{factor}}, \quad o(m) = n_m \delta_{\text{edge}}, \quad n_m = \lfloor \omega_m T_{\text{DME}} \rfloor. \quad (33)$$

Therefore

$$\frac{r_A}{r_B} = \frac{n_B}{n_A}. \quad (34)$$

Now compute n_A and n_B exactly:

$$\omega_A T_{\text{DME}} = \frac{2\sqrt{2}}{2\gamma} = \frac{\sqrt{2}}{\gamma}, \quad \omega_B T_{\text{DME}} = \frac{6\sqrt{2}}{2\gamma} = \frac{3\sqrt{2}}{\gamma}. \quad (35)$$

With $\gamma = c_2/I(2)$ from [13]:

$$\frac{\sqrt{2}}{\gamma} = 2 + \delta_{\text{edge}}, \quad \frac{3\sqrt{2}}{\gamma} = 6 + 3\delta_{\text{edge}}, \quad \delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2 = 0.012143\dots \quad (36)$$

Since $0 < \delta_{\text{edge}} < 1/3$,

$$n_A = \lfloor 2 + \delta_{\text{edge}} \rfloor = 2, \quad n_B = \lfloor 6 + 3\delta_{\text{edge}} \rfloor = 6. \quad (37)$$

Hence $r_A/r_B = n_B/n_A = 6/2 = 3$. $\square$

Remark 7.2 (Relation to the exact generation count). The ratio $|M_A|/|M_B| = 3$ has a natural interpretation: the two loop classes A and B have amplitudes in the ratio 1 : 3, matching the ratio of the mode frequencies $\omega_A : \omega_B = 1 : 3$ and matching the Standard Model generation count $d - 1 = 3$. The exactness is a consequence of the DME time quantum and the DME-Core winding condition, and is not fitted.

8 The processor T and charge bookkeeping

8.1 Definition of the processor

Definition 8.1 (Processor T). Let P_{mesh} be the orthogonal projection onto the 9-dimensional mesh resonance subspace ($\delta \neq 0$) and let P_τ be the rank-1 projection onto the unique mode with zero-state amplitude $s = +1$ (τ). The processor is

$$T = P_{\text{mesh}} + P_\tau = I - P_{\text{sub}} + P_\tau. \quad (38)$$

The F/B split is

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad B = \{\delta = 0\} \setminus \{s = +1\}. \quad (39)$$

8.2 Properties

Theorem 8.2 (Processor properties). T is a rank-10 orthogonal projection:

$$T^2 = T, \quad T^\dagger = T, \quad \text{rank}(T) = 10. \quad (40)$$

Its image is exactly the 10 Standard Model fermions $F = \{\nu, e, u, d, s, \mu, c, b, t, \tau\}$ and its kernel is exactly the 4 Standard Model bosons $B = \{\text{dark}, W, Z, H\}$.

Proof. $T^2 = T$ and $T^\dagger = T$ follow from the orthogonality of P_{mesh} and P_τ . The rank-10 property follows from $9 + 1 = 10$ for the ranks of P_{mesh} and P_τ . The identification of the image with the fermions and the kernel with the bosons is established in [6, 17]. $\square$

8.3 Charge bookkeeping

Theorem 8.3 (Charge bookkeeping). The charges satisfy

$$\sum_{j \in \text{image}(T)} Q_j = -2, \quad \sum_{j \in \text{ker}(T)} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1. \quad (41)$$

Proof. Direct computation using the charge assignment of [7]. The image of T is the 10 fermions; the kernel of T is the 4 bosons. The sum over the image is -2 and the sum over the kernel is $+1$; the total is -1 . $\square$

Remark 8.4. The framework's decomposition into outer mesh ($+2/3$, 6 modes) and middle mesh ($-5/3$, 8 modes) uses support classes. The processor decomposition uses the projector T . Both sum to -1 , matching the vacuum charge invariant.

9 The HCSM Motion Theorem

Theorem 9.1 (HCSM Motion Theorem). Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the substrate with the 14-mode multiplet $\mathcal{K}_{14}$ at $\lambda = 4$. Let $H_{\text{wind}} = i \sin((\pi/8)M)$ be the winding generator. Then:

1. H_{wind} is the unique Hermitian generator on $\mathbb{C}^{14}$ determined by the substrate and the spectral constraint $\text{spec}(H_{\text{wind}}) = \{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$. Its characteristic polynomial is $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.

2. The standing-wave amplitudes are exactly the rationals $28/9, 7/18, 21/16$ for supports $64, 48, 32$, derived from the complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.
3. The five layers $F\text{DLM}/W\text{IN} \rightarrow D\text{ME} \rightarrow \text{Core} \rightarrow E\text{FT} \rightarrow \text{Processor}$ have well-defined state spaces and unique operators.
4. The loop operator $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ is diagonal, with closed-form eigenvalue magnitudes

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}, \quad |M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho)T_{\text{beat}}/2}.$$

5. $|M_A|/|M_B| = 3$ exactly.
6. The charge bookkeeping is $\sum Q_{\text{image}} = -2, \sum Q_{\text{kernel}} = +1$, total -1 .
7. The loop is dissipative: $|M_j| < 1$ for all j .

Proof. Parts (1)–(7) are established in Theorems 3.4, 4.3, 6.2, 7.1, 8.2, and 8.3 above. The uniqueness of H_{wind} is Corollary 5.3. The dissipativity follows from $|w+1|/\delta_{\text{edge}} = 1.1574\dots$ and $e^{-(\zeta H/\rho)T_{\text{beat}}/2} < 1$, giving $|M_A| = 0.5757\dots$ and $|M_B| = 0.1919\dots$, both < 1 . $\square$

10 Status, disclosed inputs, and open items

10.1 Derived

1. The winding generator H_{wind} and its spectrum (Theorem 3.4, Corollary 3.5).
2. The trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$ (Lemma 3.3).
3. The characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ (Theorem 3.4).
4. The three D_4 orbits and the frequency-class partition (Section 3).
5. The standing-wave amplitudes $28/9, 7/18, 21/16$ (Theorem 4.3).
6. The Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$ (Theorem 4.2).
7. The uniqueness of the generator (Corollary 5.3).
8. The processor T and its properties (Theorem 8.2).
9. The exact ratio $|M_A|/|M_B| = 3$ (Theorem 7.1).
10. The charge bookkeeping (Theorem 8.3).

10.2 Disclosed inputs

1. The DME stability coefficient $\gamma = c_2/I(2)$ and time quantum $T_{\text{DME}} = 1/(2\gamma)$ from HCSM-54 [13]. In HCSM-54, γ is derived from the AGM closed form $I(2) = 1 - (1, \sqrt{2})/2$ and the closed form of $c_2 = 2I''(2)$.
2. The EFT equation of state $w = -1 - (1 - \gamma)^2/(2\pi)$, bulk viscosity $\zeta H/\rho = (1 - \gamma)^2/(6\pi)$, and Proca mass m_s from HCSM-56 [14].
3. The physical normalization $M_P^2 = Nk_L^2\lambda_L^2$ from HCSM-48 [10].

10.3 Open

1. **The loop pass duration selection.** Both T_{DME} (DME relaxation time) and T_{beat} (Level-1 internal period) are dimensionless time-like quantities. The transition rule uses T_{DME} ; Level 1 uses T_{beat} internally. The framework identifies the loop pass duration with T_{DME} by the transition rule, but does not derive from a deeper principle that T_{DME} and not T_{beat} is the physical pass duration. The ratio $T_{\text{beat}}/T_{\text{DME}} = 3.1226\dots$ is a structural constant. This is the subject of HCSM-21.
2. **The full nonlinear Einstein equations.** The composite-operator effective action with nine structural zeros is the subject of HCSM-50.
3. **The physical conversion.** The framework measures time in units of T_{DME} ; conversion to seconds requires a calibration choice external to the framework.

11 Falsifiable predictions

1. **Loop eigenvalue ratio.** $|M_A|/|M_B| = 3$ exactly. If a future measurement or computation of the framework's two-tier amplitude hierarchy yields a ratio other than 3, the Motion Theorem is falsified.
2. **Standing-wave amplitudes.** $\text{ptp}\langle x \rangle_{64} = 28/9$, $\text{ptp}\langle x \rangle_{48} = 7/18$, $\text{ptp}\langle x \rangle_{32} = 21/16$, exactly. These are rationals; any deviation would falsify the substrate algebra.
3. **Zero drift at the beat period.** $\langle x \rangle_j(T_{\text{beat}}) = \langle x \rangle_j(0)$ exactly for all j .
4. **Dissipativity.** $|M_j| < 1$ for all j ; under iteration, $M_{\text{loop}}^k \rightarrow 0$.

12 Conclusion

We have derived the motion of particles in the Hodge Complex Standard Model from the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. Motion is the composition of five layers of the substrate, each with a well-defined state space and a unique operator. Every quantity in the loop is derived analytically: the characteristic polynomial of H_{wind} , the standing-wave amplitudes $28/9, 7/18, 21/16$, the closed-form loop eigenvalue magnitudes, and the exact ratio $|M_A|/|M_B| = 3$.

The winding generator $H_{\text{wind}} = i \sin((\pi/8)M)$ is the unique Hermitian generator on $\mathbb{C}^{14}$ determined by the substrate and the spectral constraint $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$. Its characteristic polynomial is $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.

The exact ratio $|M_A|/|M_B| = 3 = d - 1$ is a consequence of the DME time quantum and the DME-Core winding condition, and matches the Standard Model generation count. All framework constants cancel.

The processor $T = P_{\text{mesh}} + P_{\tau}$ is the unique rank-10 orthogonal projection producing the Standard Model fermion/boson split. Its image is exactly the 10 fermions; its kernel is exactly the 4 bosons. The charge bookkeeping sums to -2 , $+1$, and -1 respectively.

The present paper follows the HCSM-06/07 pattern: it derives everything in its own domain that can be derived, discloses the three physical inputs (γ , EFT parameters, M_P^2) explicitly, and names the specific open items (loop pass duration, nonlinear Einstein equations) that future papers must close. The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-05: Orbit Decomposition and Inverse Participation*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-13: The Processor T and the F/B -Class Correspondence*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-24: The Charge Theorem*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-45: Graviton as Self-Paired Curvature Mode*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-46: Graviton Exact Masslessness*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-48: Faddeev–Popov G_N Normalization*, HCSM White Paper Series (2026).
- [11] S. Preschutti, *HCSM-49: Completion of the Newton Constant Sector*, HCSM White Paper Series (2026).
- [12] S. Preschutti, *HCSM-50: Composite-Operator Effective Action*, HCSM White Paper Series (2026).
- [13] S. Preschutti, *HCSM-54: Vacuum Stability and the DME Stability Coefficient*, HCSM White Paper Series (2026).
- [14] S. Preschutti, *HCSM-56: Vacuum EFT*, HCSM White Paper Series (2026).
- [15] S. Preschutti, *The WIN Motion Theorem, Version 2*, WIN Paradigm White Paper (2026).
- [16] S. Preschutti, *The WIN Time Theorem, Version 2*, WIN Paradigm White Paper (2026).
- [17] S. Preschutti, *The WIN Vacuum as a Calibration-Based Processor*, WIN Paradigm White Paper (2026).
- [18] J. M. Borwein and P. B. Borwein, *Pi and the AGM*, Wiley (1987).

A Reproducible verification

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-20_verification`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the 14-mode multiplet, the winding generator, and the loop operator, and verifies:

1. The 14-fold degeneracy at $\lambda = 4$.
2. The trace identities $\text{tr}(H_{\text{wind}}^2) = 80$ and $\text{tr}(H_{\text{wind}}^4) = 2176$ (Lemma 3.3).
3. The characteristic polynomial $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ and the spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$ (Theorem 3.4).
4. The three frequency classes and their multiplicities 3, 7, 4.
5. The standing-wave amplitudes 28/9, 7/18, 21/16 (Theorem 4.3).
6. The Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$ (Theorem 4.2).
7. The uniqueness of the generator (Corollary 5.3).
8. The processor T and its image/kernel (Theorem 8.2).
9. The exact ratio $|M_A|/|M_B| = 3$ (Theorem 7.1).
10. The charge bookkeeping (Theorem 8.3).
11. The dissipativity $|M_j| < 1$ for all j .

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

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 $H_{\text{wind}} = i \sin((\pi/8) M)$  has the characteristic polynomial
 $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$ . The standing-wave
amplitudes are exactly 28/9, 7/18, 21/16 for the three D4 orbits.
The loop operator is diagonal with closed-form eigenvalue magnitudes
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giving $|M_A|/|M_B| = 3$ exactly.
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B.2 Schema.org JSON-LD

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B.3 BibTeX for Automated Indexing

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    amplitudes are exactly  $28/9$ ,  $7/18$ ,  $21/16$ . The loop operator is
    diagonal with closed-form eigenvalue magnitudes giving
 $|M_A|/|M_B| = 3$  exactly.}
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